

Can the Galilean ideal of generative linguistics be sustained?

Abstract

The advent of large language models (LLMs) has precipitated intense critiques against the generative linguistics paradigm. While generative linguistics turns its focus inward to model the idealized computational faculty of the mind, LLM research posits that language, as manifested in use, is best understood as the probabilistic outcome of complex processing mechanisms operating on vast datasets. This controversy raises a pivotal question: in dialogue with this new generation of statistically driven LLMs, might the Galilean ideal of generative linguistics---with its emphasis on finding the most concise computational principle of our language faculty---be superseded or even rendered obsolete? I address three fundamental issues: (1) While generative theory seeks perfect configurations of mental states that necessarily transcend surface language use, what substantive value does such idealized abstract modeling hold? Can it genuinely illuminate the Galilean enigma of human cognition? (2) During the process of idealization, how should Galilean generative linguistics delineate its resilience boundaries to avoid degenerating into an unfalsifiable, self-referential system? (3) To what extent does a constructive dialogue remain possible between Galilean linguistics and LLMs?

Keywords: generative linguistics; Galilean ideal; large language models; linguistic philosophy

1. Introduction

Regarding the fundamental nature of language, Chomsky's lifelong inquiry has engaged three pivotal questions (Jenkins, 2000): Humboldt's problem (What constitutes linguistic knowledge?), Plato's problem (How is linguistic knowledge acquired?), and Descartes' problem (How is linguistic knowledge deployed?). By distinguishing between I-language (internal language) and E-language (external language), Chomsky (1986) captures the computational dimension of a biologically rooted language system. He maintains that investigating the essence necessitates uncovering the interplay among its intrinsic properties, acquisition, and usage. The first core question of biolinguistics answers that linguistic knowledge represents the operational state of the mind/brain in a speaker who "knows a particular language". The revolutionary Chomskyan shift lies in rejecting the static view of structuralism. It reconceptualizes linguistic knowledge not as a mere inventory of rules, facts, and symbols, nor as Bloomfield's "totality of utterances in a speech community" (Chomsky, 1986), but as a system of dynamic¹ mental operations.

Drawing on Plato's doctrine of anamnesis, according to which learning consists in recollecting latent knowledge from a pre-empirical realm (Demir, 2025), the generative inquiry reframes Plato's problem. It asks how the pre-structured computational capacity of the mind manifests itself upon exposure to fragmentary experience (Chomsky, 2000). This shift transforms the question of

¹ Within this framework, "dynamic" refers not to real-time processing, but to the abstract, generative nature of the computations that constitute the language faculty.

acquisition from one of inductive learning to one of cognitive triggering: experience does not instruct the mind, but rather acts as a trigger for the crystallization of a logically prior system. The foundational issue, therefore, is the unfolding of a biologically constrained, formal schema that is, in a Platonic sense, already implicitly present and awaiting actualization.

The deployment of linguistic knowledge constitutes a dynamic cognitive realization process. From this perspective, a speech act is not merely the production of a string of symbols, but the procedural actualization of thought via syntactic computation. This actualization operates through three interconnected domains. On a cognitive level, the core property of the internal language faculty (I-language) is its recursive generativity, which allows a finite set of mental operations to yield a discrete infinity of expressions. On a communicative level, the internal structures must be externalized for use, a process that must conform to the linearization constraints of the sensory-motor systems (e.g. producing speech sounds or signs in a sequence). On an evolutionary or historical level, language change across time arises from the subtle reconfiguration of underlying grammatical options (parameters) driven by accumulated language use across generations. Thus, studying externalized use must transcend surface forms to examine generative dynamics and realizational constraints.

In addressing Humboldt's, Plato's, and Descartes' problems, Chomsky adopts a distinctly Galilean approach (Allott et al., 2021; Collins, 2023), expressly rejecting Karl Popper's falsificationism in the philosophy of science (Helfenbein & DeSalle, 2005; Rescher, 2000). The "Galilean approach", given its modern formulation by nuclear physicist Weinberg (2001), draws inspiration from Galileo's scientific practice. It holds that physicists should seek mathematically elegant models of the cosmos, even when most empirical phenomena remain unexplained. The Galilean philosophy of generative linguistics refers to the pursuit of a deep understanding of language faculty through idealized abstract modeling of mental or brain states by deliberately disregarding interfering phenomena. Its hypotheses require continual adjustment via empirical verification. For Chomsky, Galilean science means pursuing depth of understanding over breadth of coverage of facts, with ideal theories being those achieved through Galileo-Newtonian rigor (Chomsky et al., 2002). Researchers must avoid being misled by theory-challenging exceptions, remembering that reality often contradicts commonsense intuition.

Among Chomsky's tiers of theoretical success---observational (covering the data), descriptive (characterizing I-language), and explanatory---the last represents the Galilean ambition of the field. It offers insights about why the language faculty must have the computational properties it does (Chomsky, 1965). Notably, at his 2024 Ulysses Medal acceptance at University College Dublin, deep learning pioneer Geoffrey Hinton publicly repudiated Chomsky's scientific philosophy, arguing that large language models (LLMs)---starting from random weights and massive data---can effectively simulate natural language. This contrast epitomizes a fundamental schism: the former seeks to explain the necessity of the language faculty from first principles, while the latter demonstrates that surface-level usage can emerge from statistical approximation, without recourse to a theory of the underlying faculty.

In recent years, the rapid ascent of LLMs has brought this methodological tension into sharp relief. Given that these models achieve human-level fluency and comprehension by leveraging massive data and statistical approximation, a pressing question emerges: does Galilean linguistics or do

LLMs hold greater persuasive power and offer a closer approximation to the true character of language in the ongoing debate (Martínez et al., 2024; Fox & Katzir, 2024)? Criticism about generative linguistics argues that the success of data-driven models obviates the need for a priori, generative theories of a specialized language faculty (e.g. Piantadosi, 2024). It is against this contemporary backdrop that the following fundamental questions must be addressed:

- (1) While generative linguistics seeks perfect configurations of mental states that necessarily transcend surface language use, what substantive value does such idealized abstract modeling hold? Can it genuinely illuminate the Galilean enigma of human cognition?
- (2) During the process of idealization, how should Galilean generative linguistics delineate its resilience boundaries to avoid degenerating into an unfalsifiable, self-referential system?
- (3) To what extent does a constructive dialogue remain possible between Galilean linguistics and LLMs?

The argumentative architecture of the present study constitutes a philosophical intervention at the meta-theoretical level, the significance of which is distinct from any empirical adjudication between generative linguistics and LLMs. While prior responses to critics such as Piantadosi (2024) have noted the categorical gap between LLM performance and human linguistic cognition, my analysis executes a novel modal reorientation of the entire debate. Existing contributions in this debate, exemplified by Piantadosi (2024), often assess competing frameworks by comparing their empirical-explanatory performance. Treating the empirical adequacy of LLMs under specific training conditions as a decisive standard for judging the correctness of generative linguistics commits a modal fallacy. A constitutive condition that any cognitive faculty must satisfy across all possible realizations can't be refuted by demonstrating that a particular implementation, constrained by specific historical circumstances and architectural choices, produces similar output patterns. The philosophical import of this modal distinction lies in its application to an ongoing scientific controversy, where it functions as a critical instrument that prevents a research program from being dismissed on illegitimate grounds. The study thus performs a diagnostic function. It identifies a widespread confusion that has led commentators to conflate explanatory levels that logic requires remain separate.

I further contribute to the philosophy of special sciences by developing meta-theoretical safeguards for Galilean idealization. While prior discussions (Collins, 2023) have diagnosed methodological tensions within generative linguistics between idealization and empirical adequacy, they offer no operational framework to demarcate where idealized core syntax ends and extralinguistic subsystems begin. This study fills that gap by introducing three interlinked methodological constructs: the domain of idealization (the bounded set of phenomena the core theory aims to account for), extra-domain necessary subsystems (cognitive components abstracted away in idealization yet required for real-world linguistic behaviour), and parameterized interface couplings (modelled interaction mechanisms linking core syntax to those external cognitive systems). I additionally deploy minimum-description-length heuristics (a meta-criterion balancing model complexity and data-fitting cost) and a constrained-consistency testing standard (evaluating theories via compatible cross-paradigm predictions rather than strict one-to-one neural-isomorphism). These tools help judge when explanatory weight ought to reside in core syntactic operations versus downstream filtering, and flag when auxiliary hypotheses risk degenerating into Lakatosian protective-belt ad-hoc moves (auxiliary assumptions deployed merely to shield core claims from counter-evidence). Rather than merely rehearsing existing anti-reductionist or structural-realist

observations in biolinguistics, my accounts attempt to supply concrete, testable criteria to prevent generative theorizing from lapsing into unfalsifiability without dismissing its Galilean explanatory ambitions.

The philosophical contribution of repositioning LLMs as epistemic tools for theory evaluation lies in its reconfiguration of the concept of evidence in theory testing. Prior research (Portelance & Jasbi, 2025) has already treated LLMs as grammar-induction discovery tools for linguistic theorizing. This paper pushes this line of thought further. Under standard practice, counterevidence to a theory originates from empirical observations within the ontological domain that the theory governs. I argue that systematic deviations arising from a radically heterogeneous implementation architecture (e.g. LLMs) can equally serve as testing resources for theoretical constraints. The analysis advances existing methodological discourse by demonstrating that when an implementation path fails to reproduce the regularities predicted by a theory, this mismatch may offer meta-theoretical grounds for assessing the proper boundaries of idealization. This logic effectively leverages alternative realization as an epistemic lever for theory evaluation, transforming the appropriateness of idealization from an untestable presupposition into an object that can be scrutinized across heterogeneous realization spaces. The argument thereby introduces a novel type of evidence into scientific methodology, one that does not necessarily seek to confirm or refute the empirical content of a theory in the first instance, but rather constrains the scope of reasonableness of the idealization strategy of a theory. This methodological innovation carries general referential value for many higher-level sciences that rely on idealized models.

2. In pursuit of perfect mental configurations: epistemological implications

Since its modern inception, generative theory has sought to construct an idealized model of the language faculty. This endeavor was crystallized in the principles and parameters framework, which reduced the vast complexity of human language phenomena to an operable and analyzable system of universals and parameters. With the advent of the minimalist program, the theory transcended universal grammar, positing that language must be perfectly designed. Its mechanisms operate in the most economical manner possible while satisfying only the most basic interface conditions (namely the conceptual-intentional and sensory-motor systems) (Chomsky, 1995). Departing from earlier generative assumptions, the minimalist program contends that a “perfect” language would require no substantive, specialized linguistic engine in the brain. The strong minimalist thesis goes further, suggesting that, in an ideal scenario, universal grammar would contain no substantive language-specific rules, though it still posits merge as a proprietary computational operation, rendering the language faculty perfectly designed (Golumbia, 2010). In other words, no superfluous structures or operations would exist; all syntactic generation would serve solely to meet interface demands with maximal efficiency.

From a philosophy-of-science perspective, I present that the pursuit of perfect mental configurations is best understood not as a straightforward form of classical reductionism, but as a form of rationalist inquiry that employs Galilean idealization. This approach shares with reductionism a commitment to uncovering fundamental principles by abstracting away from surface complexity. However, their epistemic aims diverge crucially. Classical reductionism (Gillett, 2016; Nagashima,

2014), as in physics, typically aims for its idealized models to serve as a foundational core from which the full breadth of empirical phenomena can, in principle, be reconstructed through successive approximation (Sachse, 2007). In contrast, the Galilean style in linguistics does not seek to reconstruct all possible linguistic facts from the bottom up. Instead, its idealization is more radical and its goal more circumscribed: to isolate and characterize the computational essence of the language faculty as an autonomous system. While such an operation places formal constraints on what is possible, it does not fully determine the full range of observable linguistic variability. The value of the model lies not in its capacity for comprehensive reconstruction, but in its power to reveal the necessary and sufficient conditions for the system to exist. I contend that while this method allows us to abstract language into minimalist principles, we can't expect such a purified model to fully reconstruct the entirety of linguistic facts.²

This circumscribed goal finds a sound justification in the broader landscape of science. The limitations of the reductionist program were famously crystallized in Philip Anderson's (1972) seminal thesis *More Is Different*. Anderson argues that even if complex phenomena supervene on lower-level constituents, fundamental laws governing those constituents cannot by themselves fully explain higher-order complexity. Novel explanatory principles and systemic properties emerge at higher organizational levels, and aggregate-level behavior cannot be straightforwardly extrapolated from the properties of individual components. This hierarchical view of science, in which emergence demands level-specific explanatory frameworks, lends support to the methodological stance of generative linguistics. On this view, linguistic phenomena may not be exhaustively explained by appealing solely to general-purpose cognitive mechanisms or statistical inference. Language could form a distinct organizational level within cognition, whose account may require its own foundational principles, which is the core aim of Galilean linguistics.

Given that LLMs increasingly approximate human-level language generation and comprehension (Piantadosi, 2024), one might question the value of generative linguistics, which can't reconstruct the full breadth of real-world data. Yet history shows that revolutionary theories often begin with radical simplifications. Economic models assume a homo economicus (Hollis & Nell, 1975); Newtonian gravity ignores air resistance; special relativity presupposes ideal inertial frames; quantum mechanics even relies on the counterintuitive "observer effect". These theories penetrate the surface of reality precisely because they refuse to be shackled by empirical noise. Generative linguistics follows the same epistemic strategy: by hypothesizing an ideal speaker-hearer, it deliberately strips away the noise, variation, and accidents of usage to reveal the logical boundaries of the language faculty. The value of generative linguistics lies in its ability to liberate language from this patchwork, exposing its core computational operation---recursive merge---through idealization. Without such abstraction, we risk drowning in phenomena, unable to distinguish signal from noise. Recognizing this, the onus is on generative theory to justify its idealization boundaries empirically, not merely stipulate them.

However, a subtle relation remains between idealized models and biological reality. While the merge operation is mathematically elegant, neuroscience has yet to identify dedicated neural circuits corresponding to it. Structural realism in the philosophy of science asserts that the value of a theory lies not in whether its ontological commitments are empirically verified at the outset, but in whether it captures the relational structure of phenomena, reveals the intrinsic constraints of logical

² It should be clarified that this is not to criticize generative linguistics for violating this principle.

systems, and provides explanatory power unattainable through other means (Landry & Rickles, 2012). For instance, Maxwell's equations initially served as mere mathematical descriptions of electromagnetic phenomena, only later receiving physical interpretations (Maxworth, 2024). Similarly, not every linguistic principle may map directly onto dedicated brain circuits, yet collectively, they elucidate the computational constraints that language faculty must satisfy. The curious inquiry is whether this approach provides an indispensable form of explanation or constitutes a Procrustean idealization that fails to advance our understanding of the actual cognitive system.

The curious inquiry can be reframed by distinguishing between different causal-explanatory roles. Ross and Bassett (2024) argue that in neuroscience, mechanism is often used in overly broad ways that risk vacuity, and they propose the concept of an enabling constraint, a causal factor that does not directly produce an outcome but instead shapes the possibility space within which other mechanisms operate. Generative linguists' postulation of computational efficiency (e.g. minimal search, see Krivochen, 2023) can be understood as precisely this kind of enabling constraint. Rather than claiming to identify biophysical mechanisms that drive neural firing, it specifies the boundary conditions that any viable neurobiological implementation of syntax must respect. From this perspective, the absence of dedicated neural circuits for merge suggests that we might be looking at the wrong explanatory level.

In recent years, a growing body of research has begun leveraging phase coherence of neural oscillations to investigate the mechanisms underlying syntactic parsing, at least attempting to disentangle core linguistic modules from general cognitive processes, though the feasibility of such separation remains uncertain (Bai et al., 2022; Calmus et al., 2020; Martin, 2020; Murphy, 2025). This approach seeks to align the temporal dynamics of neural oscillatory signaling with the hierarchical construction of linguistic structure. Phase refers to the position of an oscillatory signal within its periodic waveform cycle. Phase coherence, in turn, quantifies the degree of phase synchronization between oscillatory activities across distinct neuronal populations, serving as a critical metric for assessing coordinated neural activity. Phase synchronization manifests in multiple forms, including within-frequency synchronization across different brain regions (Fries, 2005) and cross-frequency coupling, where oscillations of distinct bands exhibit phase-dependent interactions. During syntactic parsing, neural oscillations---both within and across frequency bands---demonstrate varying degrees of phase coherence. Thus, phase coherence reflects the integration of linguistic representations across hierarchical levels, operating at distinct temporal scales (Giraud & Poeppel, 2012).

A recent line of neurophysiological research about neural oscillations has found that low-frequency bands (e.g. δ -band) correlate with the hierarchical construction of syntactic structure, while high-frequency bands carry lower-level sensory information (Bai et al., 2022; Ding et al., 2016). Such findings suggest that the abstract, hierarchical representations posited by generative linguistics may have correlates in neural dynamics. Crucially, however, these correlations fall short of establishing fine-grained causal isomorphism between syntactic operations and neural mechanisms. As shown in **Figure 1**, while phase-coupled oscillatory patterns align with dimensions of syntactic trees (Marcolli et al., 2023), the mapping remains non-isomorphic and nonlinear (Kazanina & Tavano, 2023).

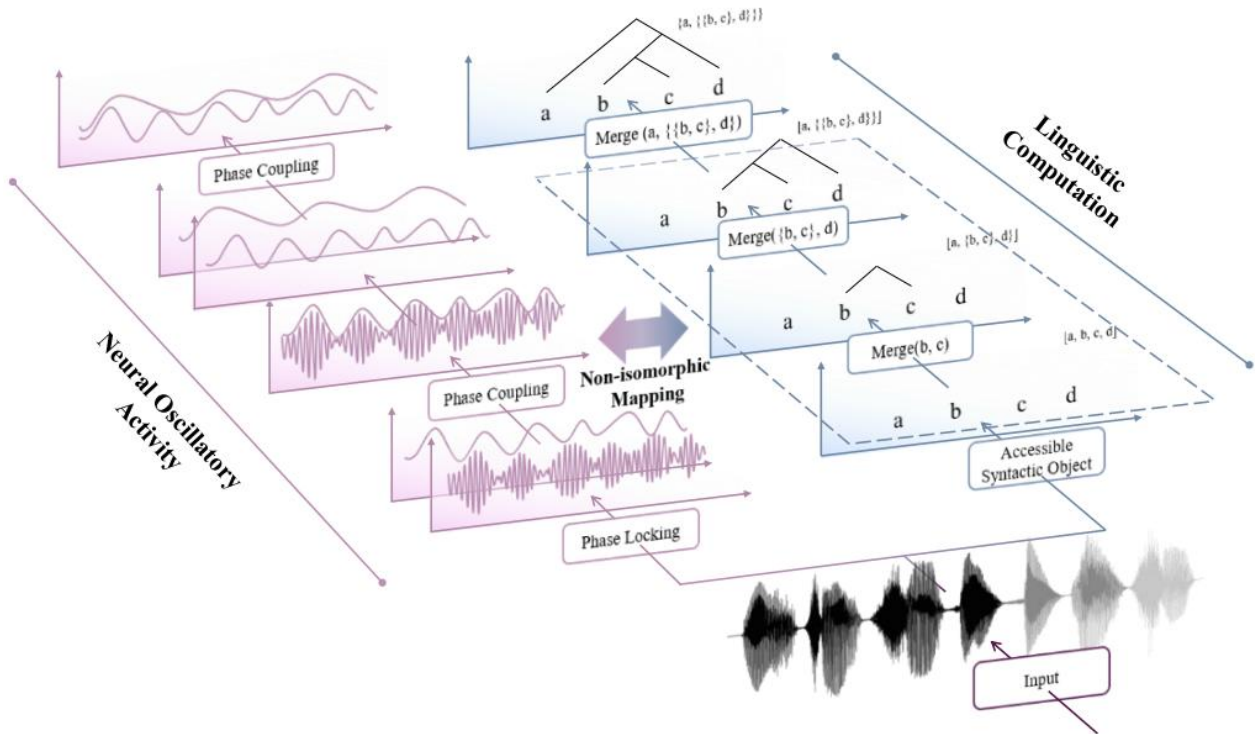


Figure 1. Syntactic operations and their neural correlates (Qi et al., 2025: Figure 1, p. 300)

More advanced models have moved beyond merely seeking temporal alignment between oscillatory dynamics and syntactic structures. Murphy (2025) proposes a concrete mechanistic link: low-frequency δ oscillations provide a phase code for hierarchical phrase structure building, θ - γ phase-amplitude coupling assembles lexical feature bundles into morpheme-sized units, and δ - θ coupling determines headedness through competitive modulation of coupling strength. Crucially, the ROSE (representation, operation, structure, encoding) architecture demonstrates how the non-associativity of syntactic merge, namely the property that $(a \circ b) \circ c \neq a \circ (b \circ c)$, can be formally instantiated by neural oscillations: each merge operation occupies a distinct δ -phase segment, with a β -band commit burst freezing the composite unit before the next merge, thereby preserving hierarchical history. This goes beyond simple temporal alignment by showing how the algebraic properties of syntax (e.g. the magma structure of merge) can be neurally implemented via cross-frequency coupling mechanisms that are both biologically plausible and computationally explicit (see also Marcolli & Berwick, 2025, for a mathematical formalization of phase synchronization linking directly to algebraic models of syntax).

Based upon the preceding discussion, I now offer a rebuttal to recent literature. Piantadosi's (2024) argument entails a crucial philosophical leap. It translates the engineering success of language models in fitting linguistic phenomena into an empirical falsification of the core methodology of generative linguistics and its theoretical constructs (such as the poverty of stimulus). His logic is that if a relatively unconstrained statistical model can rediscover grammar from data, then the language faculty need not be presupposed as a highly specialized computational system. The essence of this leap is to take the empirical adequacy of one type of implementational model (how linguistic behavior is realized from data) as the ultimate yardstick for judging the theoretical necessity of another type of constitutive theory (what linguistic knowledge can and can't be like).

To situate my position, I consider Katzir's (2023) critique of Piantadosi (2024), which

systematically exposes three fundamental flaws of LLMs as theories of human linguistic cognition. Regarding inductive bias, LLMs fail to acquire fundamental syntactic universals, such as the Ross constraints, from data volumes comparable to those available to children, revealing that their learning mechanisms are fundamentally different from the innate biases of the human language faculty. Concerning typological universals, the architecture of LLMs can't naturally give rise to cross-linguistically shared constraints (e.g. island conditions, the finite-state nature of phonological rules), which is key evidence used by generative linguistics to argue for the internal constitutive form of the language faculty. On the competence-performance distinction, failures in LLMs are directly attributable to knowledge gaps, whereas human failures often stem from real-time processing resource limitations; this fundamental difference in nature indicates that LLMs completely conflate the two analytical levels of linguistic competence and performance. Although Katzir's argument is powerful, his critique remains primarily anchored to empirical model failures but has further potential to fully elevate it to the level of philosophy of science.

I introduce a custom-defined modal distinction labelled accidental truth and necessary truth, not to be read as standard modal-logical truth-predicates. The engineering success of LLMs yields condition-dependent surface-level empirical adequacy. LLM performance emerges from historical corpora, silicon-based architectural constraints, and cultural-pragmatic distributions. All its linguistic outputs derive from statistical traces of empirically observed language. Alter these boundary conditions, and its linguistic performance may shift substantially. Such conditional performance demonstrates only one possible way to reproduce human-like surface linguistic patterns within our actual world, rather than capturing the inherent computational form of the language faculty. By contrast, generative linguistics targets the nomologically necessary constitutive principles of the language faculty: computational constraints that hold across viable realizations of this biological cognitive system. Piantadosi's (2024) fallacy lies in mistaking the condition-dependent empirical adequacy of one implementational model for a decisive refutation of a constitutive theory investigating these nomologically necessary constraints. This category error attempts to use the answer to "how possible" to negate the question of "why necessary". It substitutes the existence proof of a statistical model for the philosophical and scientific inquiry into the internal constitutive form of the biological mind, effectively dissolving the deep interrogation of cognitive ontology with engineering success. The success of LLMs fails to demonstrate that they are theoretically superior or necessary for explaining the efficiency of language acquisition and the identity of its final state within the limited and noisy environment of human learning.

However, even if generative linguistics is irreplaceable in its quest for cognitively necessary truth, the challenges and skepticism it faces are intensifying in the contemporary academic context. Addressing this, Hao (2025), from a history of science perspective, suggests that this crisis is a consequence of the generative school's overly limited social ambitions. The predicament of the generative school lies in its focus on seeking internal theoretical explanatory power while neglecting to build broad external social connections, which represents a closed academic culture. In contrast, the success of LLMs lies precisely in their engineering and socialization, successfully attracting the investment and attention of diverse stakeholders. Therefore, the way forward for generative linguistics is not to intellectually defeat the models, but to expand its social ambitions, translating theoretical achievements into broader social value, thereby making the external world a stakeholder in its success.

I counter that Hao's prescription, grounded in the social nature of scientific success, itself harbors a profound philosophical paradox. It optimistically assumes that core generative theories can be effectively translated or transferred into the external world. The field where the social value Hao calls for is generated and transmitted is precisely the practical-social world that generative linguistics brackets, a world where meaning is constituted by use, understanding depends on context, and value is manifested in solving specific, contingent practical problems. When a theory dedicated to exploring necessity attempts to enter this social system of value exchange based upon contingency, generative linguistics faces a profound paradox: its theoretical power derives precisely from its detachment from specific social contexts, yet the realization of social value requires its re-embedding into those very contexts. I posit that this constitutes an ontological transgression. To become socially valuable, the theory must translate its precise, self-consistent formal representations into narratives understandable and deemed useful by the public, corporations, or governments. This process inevitably entails the dissipation of theoretical rigor and the simplification of core claims. Therefore, the perceived insularity of generative linguistics, viewed from this perspective, is a form of theoretical steadfastness. Hao invokes the history of science to demonstrate the success of social translation. Pasteur's germ theory succeeded because the theoretical entity of microbes could transition relatively stably across different contexts---laboratory, medical practice, public health policy---without losing its core identity. Although generative linguistics has been indirectly integrated into technology (e.g. providing foundations for compiler parsing implementations, see Krithivasan, 2009), the current wave represented by LLMs demonstrates a form of value that generates immediate economic and social interaction. The theoretical constructs of generative linguistics can't maintain their identity after socialization, because mathematically necessary constraints risk becoming commodities tailored for maximal social utility. Such a paradigm is inevitably isolated in a society that prioritizes immediate utility and instrumental rationality.

3. Revisiting falsifiability: the resilience boundaries of generative theory

Having rebutted several recent studies, has the current Galilean linguistic paradigm reached its maximum potential value? I concur with some points raised by Collins (2023). In physics, idealization (e.g. an ideal pendulum) remains a model concerning the behavior of an external object, merely stripped of interfering factors. However, in Galilean linguistics, when we abstract away all external and internal interfering factors (such as memory limitations, prosody), what we obtain is not an ideal language user, but merely a specific component within their mind. Consequently, idealization in linguistics differs in nature from that in physics; it potentially abstracts away precisely those other cognitive systems essential for language, rendering the ideal speaker-hearer an inappropriate target for Galilean idealization. Collins concludes that the Galilean style label is both appropriate and misleading for generative linguistics. It is appropriate in its pursuit of abstract and mathematical explanation; misleading because its specific application must be modified due to the fundamental difference in the object of study (internal mental states vs. external kinematics).

Building upon Collins' view, I further propose that if Galilean linguistics treats its mathematical constructs as theoretical components with well-defined domains and measurable interfaces, it still holds the potential to approach the level of abstraction akin to physics in the future. I introduce two

concepts: the domain of idealization and extra-domain necessary subsystems.³ The former refers to the set of phenomena and boundary conditions that generative theory explicitly claims to explain, such as the applicable variables, time scales, and causal levels. The latter refers to other components temporarily abstracted away by generative theory but indispensable in the actual cognitive system that influence observed outcomes, such as long-term memory retrieval processes, response biases induced by emotional states, and posterior constraints generated by pragmatic inference. Operationalizing the domain of idealization can prevent generative theory from unconsciously overstepping its bounds, namely generalizing conclusions valid only within a specific domain into claims about the ideal speaker-hearer.

Therefore, I advocate for the explicit formulation of interfaces. Here, I define an interface as a parameterized coupling mechanism between the formal computational component and external systems such as memory, attention, and inhibitory control.⁴ Further research can formalize the mapping from extra-domain subsystems to observed behavior as parametric functions. So-called interface parameters are mathematical objects that couple the abstract core with external processes. To illustrate, consider the formalization of merge. Zyman (2024) contrasts set-theoretic definitions of merge (where syntactic objects are sets, like {to, her}) with graph-theoretic definitions (where syntactic objects are phrase-structure trees). Suppose the core formal component adopts a graph-theoretic merge that obeys the no-tampering condition (NTC), under which merge can't delete or modify existing syntactic objects. An interface parameter can then modulate whether this NTC-obeying merge is always enforced or whether, under high working memory load (e.g. when the number of open syntactic dependencies exceeds a threshold such as $n > 4$), the parser defaults to a memory-saving set-theoretic compression that treats {to, her} as an unordered set. This parameter generates a testable prediction: under high memory load, speakers should temporarily accept structures that violate NTC. Interfaces, including such parameters, are part of the theoretical statement. Generative theory can declare the ontological meaning of these parameters, their expected sign direction and measurable scale, and provide a clear quantitative formulation of how they modulate the core generative process. Thus, when we state that the abstracted object is a component, we simultaneously answer the question of how this component relates to other elements within the real cognitive ecology.

My illustration highlights that theorists possess two distinct degrees of freedom for building generative accounts. One option works outward-facing: we hold the core syntactic operation fixed and deploy interface parameters to capture behavioral deviations, as demonstrated in the NTC example above. A second option works inward-facing: we may instead revise the internal formal properties of merge, the foundational structure-building operation of syntax. This distinction raises a critical meta-theoretical trade-off concerning competing formal definitions of merge. A bare, unconstrained merge, the conception where any two syntactic objects may be combined without built-in restrictions, keeps the syntactic core very compact, yet it generates a vast space of

³ Here, “domain of idealization” and “extra-domain necessary subsystems” are methodological constructs used to delineate the scope and interfaces of a theoretical model, distinguishing it from the ontological distinction between the faculty of language-narrow sense (FLN) and the faculty of language-broad sense (FLB). The FLN/FLB distinction (Chomsky, 2015; Hauser et al. 2002) classifies cognitive capacities: FLN refers to the computational core of language (e.g. recursive merge), arguably unique to humans and unique to language, while FLB encompasses this core plus other broader cognitive systems used in language (e.g. working memory, inhibitory control).

⁴ This operational definition of “interface” as a parameterized coupling mechanism is distinct from the theoretical, modular interfaces (e.g. sensory-motor system, conceptual-intentional system) postulated in generative linguistics. The former is a methodological tool for modeling interactions between a core computational component and external systems, whereas the latter refers to fundamental, fixed components of the language architecture.

ill-formed syntactic objects that must later be filtered away by post-syntactic filtering layers (the conceptual-intentional and sensory-motor output layers, distinct from my custom-defined interfaces). From a minimum description length perspective (Grünwald, 2007), theoretical complexity is not a zero-sum game of merely shifting burdens between core syntax and these post-syntactic filtering layers. Adding modest constraints within merge, as pioneered in feature-driven minimalist grammar (Stabler, 1997: a framework that embeds licensing conditions inside merge rather than relying solely on post-hoc post-syntactic filtering), slightly enlarges the descriptive length of the core syntactic model. Nevertheless, such built-in restrictions can prune entire families of illicit derivations in one go, drastically cutting down the extra descriptive work that would otherwise have to be undertaken by these post-syntactic filtering layers. The overall total description cost, summing both model complexity and data-fitting cost, may therefore turn out lower even if the core syntax becomes less austere.

Accordingly, theoretical simplicity cannot be equated exclusively with keeping the merge operation itself as conceptually parsimonious as possible. Over-commitment to the bare-merge ideal creates the risk of treating post-syntactic filtering layers as an all-purpose dumping ground for explanatory weight. When unexplained linguistic regularities are repeatedly delegated to these late-stage filtering mechanisms without explicit, quantifiable interface-parameter formulations, the research programme drifts toward the protective-belt danger Lakatos (1978) warns against: ad-hoc auxiliary assumptions invoked only to save core commitments from contradictory evidence. This does not mean that explanations drawing upon post-syntactic filtering should be rejected outright. Rather, minimum-description-length-oriented evaluation supplies an additional heuristic: whenever we relegate complexity to late-stage post-syntactic filtering mechanisms, we ought to examine whether a modest increase in the formal complexity of core syntactic operations could deliver a better global balance of descriptive costs, before defaulting to off-loading explanatory burden onto such downstream machinery.

Having established this meta-theoretical heuristic for theory building, I now turn to empirical evaluation criteria for the resulting formal models. I emphasize a testing logic of constrained consistency, which differs from seeking isomorphic realization. As mentioned earlier, mathematical derivations do not necessarily find isomorphic mappings at the behavioral or neural levels. We should still require that generative theory attempts to maintain relative consistency across multiple independent measurement paradigms or exhibits explicable discrepancies. Generative theory should output specific shapes and relational expressions concerning the distribution of observations, rather than seeking post-hoc narratives to explain a particular measurement result only after obtaining it. For instance, treating labeling merely as a formal choice, without assuming it corresponds to any specific psychological process, allows for non-isomorphic predictions. As the number of available labels changes from unique to multiple in one measurement, the uncertainty of this rule might manifest as discretization of judgment outcomes. In another measurement, it might not cause changes in processing time, but only amplify inter-individual differences. The two types of results need not correspond numerically. However, if the abstract core produces compatible constrained predictions across different measurements, its explanatory power is enhanced; if they contradict each other, one must return to the interface assumptions to test whether the inconsistency is caused by extra-domain coupling, rather than directly attributing the contradiction to the untestability of generative linguistics.

Although Collins points out that Galilean abstraction leaves only a certain mental component, my strategy is to treat this as the starting point, not the endpoint, of theoretical responsibility. We can both acknowledge the limited nature of the component and demand that generative theory makes falsifiable statements about when this limited nature is no longer sufficient to explain phenomena. Specifically, theory must provide criteria for its range of applicability, namely under what conditions of cognitive load, contextual complexity, or time scale the predictions of the single component will systematically deviate from observations, thus necessitating the introduction of coupled descriptions involving additional subsystems. Incorporating such standards for when model extension is required into theory prevents retreating into an unfalsifiable protective belt or post hoc justifications when facing counterexamples. This methodological shift binds formalization to the construction of transparent interfaces: mathematical simplicity remains a value, but each concise formulation must be accompanied by arguments concerning its domain of applicability, interface functions, and testable predictions. Only when abstraction ceases to be an implicit philosophical commitment and becomes a claim with explicit boundaries and error budgets can Galilean idealization avoid degenerating into an isolated device describing merely one component of the mind.

Even with explicit criteria for the domain of applicability of a theory in place, we still confront a philosophical obstacle to falsifiability. The Duhem-Quine thesis points out that when a theoretical prediction conflicts with observations, we can't directly attribute the error to the theory itself, because the testing process depends on numerous auxiliary assumptions (Caamaño-Alegre, 2025). Thus, absolute falsification faces practical challenges in scientific inquiry. Nevertheless, the notions of riskiness and degrees of falsifiability can still provide important methodological constraints for theory evaluation (Vignero & Wenmackers, 2021). When experimental data conflict with theory, theory typically adjusts auxiliary hypotheses (e.g. parameters in externalization systems) rather than abandoning core postulates. Though criticized as a falsification-immunization strategy (Al-Mutairi, n.d.), this mirrors common practices in complex sciences: quantum field theory routinely introduces new particles to preserve the standard model. The deeper philosophical issue involves reconciling falsifiability with highly abstract theorizing. As Russell and Norvig (2022) observe that science blends facts and theories, with the latter never wholly superseding the former.

The critical inquiry should address how theories avoid degenerating into self-referential generativism when over-relying on idealizations, yet evade empirical fragmentation when over-accommodating linguistic facts. Scientific progress often emerges from dialectical tension: universalist aspirations must negotiate with particularist sensitivities when demarcating resilience boundaries. Lakatos and Patton's (2014) research program framework proves instructive: core assumptions form the basis, while auxiliary hypotheses constitute the "protective belt". Falsifying instances trigger belt adjustments rather than abandonment of the core basis, though Lakatos warns that excessively ad hoc modifications risk pseudoscientific degeneration. When myriad exceptions are attributed to interface conditions of usage externalization, explanatory power may dilute until the pure language faculty becomes empirically vacuous.

When confronting the non-configurational syntax of Australian Aboriginal languages (Austin & Bresnan, 1996; Hale, 1983), generative theory appears to reach its limits. The externalization theory of syntax offers a more flexible solution: it radically separates core syntactic operations (i.e. merge) from phonological realization, acknowledging that universal grammar may only constrain

representations at the level of logical form, while surface linear order is parametrically determined by externalization systems (Dobashi, 2019). This theoretical adjustment transforms “mutual concessions between facts and theory” into new exploratory dimensions. Much as general relativity did not negate Newtonian mechanics but delineated its boundaries, when the syntactic fission of Dyirbal reveals verbs assignable to distinct lexical categories across contexts (Dixon, 1972), generative linguistics preserves the universality of merge through root merge (Chomsky, 2013; De Belder & van Craenenbroeck, 2015) while accommodating typological variation. By redefining theoretical boundaries (faculty of language-narrow sense versus faculty of language-broad sense), it achieves observational and explanatory unity at a higher level.

Despite the above commendation, we still need to be cautious that when theories consistently lack converging evidence from neighboring disciplines (e.g. neurobiology) and more importantly, undergo excessive ad hoc modifications, they risk being viewed as terminological gameplay. I contend that auxiliary hypotheses warrant scientific progress only when they guide the justification of novel phenomena (e.g. cross-linguistic patterns); if merely accommodating known counterexamples, they epitomize degeneration. Lakatos’ (1978) research programme framework elucidates this distinction: the vitality of a theory hinges on whether its problem shifts generate excess empirical content, namely guiding the discovery of unobserved phenomena. When theoretical adjustments degrade from “productive hypothesis generators” to “explanatory patchworks”, and if interface conditions for strong/weak phase multiply merely to salvage counterexamples, their epistemic value undergoes qualitative deterioration. This risk is acute in cognitive science. While biolinguistic postulates may lack empirical constraints, their merit lies in constructing order in chaos. Yet unchecked theoretical liberty may culminate in Feyerabend’s (1988) methodological crisis of “anything goes”. For enduring relevance, generative linguistics should embrace both bold conjectures and rigorous refutations.

4. Generative linguistics and LLMs: theoretical stress-testing

As previously discussed, although Hao’s (2025) advocated social nature of scientific success methodologically leads inevitably to ontological transgression, this concern does not imply that there is a lack of potential for constructive dialogue between generative linguistics and the data-driven paradigm represented by LLMs. On the contrary, it is precisely by rejecting social visibility as the ultimate evaluative standard that such dialogue can be seriously engaged with methodologically. Portelance and Jasbi (2025) offer a constructive approach to understanding the dialogue between LLMs and generative theory. Their research suggests that LLMs, particularly structurally transparent grammar induction models, can actually be viewed as modern discovery procedures, which means computational tools capable of automatically inducing grammatical structures from linguistic data. This perspective elevates LLMs from mere behavioral simulators to potential collaborators in theory construction. They emphasize that the criteria of observational, descriptive, and explanatory adequacy, established by generative linguistics, provide a rigorous framework for evaluating LLMs as models of human cognition. LLMs currently show significant achievement in observational adequacy but retain major deficiencies in descriptive and explanatory adequacy, as they struggle to provide interpretable structural descriptions and fail to simulate the child acquisition process under limited data. This distinction between levels of adequacy precisely

sets the stage for LLMs to participate in theory testing: models must not only produce correct sentences but must also intrinsically capture the abstract structural constraints believed to constitute the core of human linguistic knowledge. Therefore, the dialogue between LLMs and generative theory should not remain at the level of superficial performance comparison, but should delve into the core question of whether the grammar internalized by the model is isomorphic to the grammar of the human mind.

Building upon Portelance and Jasbi (2025), I argue that methodological dialogue should not be based upon a clear-cut division of labor between generative theory and LLMs, but should be founded on counterexample generation and theoretical stress-testing. The value of LLMs lies in creating extreme scenarios for generative theory, thereby exposing their necessity claims to the possibility of systematic failure. As proposed earlier, what is necessarily true might in fact be unavoidable within the current formal system, not necessarily within neuro-cognitive reality. The role played by LLMs here is to provide a heterogeneous implementation path to test whether a principle proclaimed as structurally necessary is merely a byproduct of a specific theoretical modeling approach.

This conception of theoretical stress-testing finds support in the framework of computational rationality developed in recent cognitive science. As Griffiths (2026) argues, human cognition should be better understood not as logical deduction or classical Bayesian optimization, but as resource-constrained probabilistic inference: given limited time, data, and computational capacity, intelligent behavior emerges from approximate Bayesian inference strategies. From this perspective, both the human language faculty and LLMs can be viewed as different cases of inference under resource constraints, the former shaped by biological evolution and neural architecture, the latter by training objectives and architectural design choices. If a syntactic principle claimed to be necessary for human language can be derived from a resource-constrained probabilistic model without being explicitly built into its architecture, this does not automatically refute the principle. However, it does shift the burden of proof. Generative theory must then demonstrate that the principle reflects a genuine cognitive constraint rather than a mathematically elegant but contingent property of a particular idealization scheme. In this sense, the LLM serves as a computational-level benchmark, which is a probe for distinguishing necessary cognitive invariants from accidental modeling artifacts.

A concrete illustration of this stress-testing logic can be drawn from the debate over island constraints. When generative theory claims that certain island constraints can't be induced from statistical learning, these claims are provable within the generative system, but their external validation remains contentious (Snyder, 2000). LLMs offer a resource: although their internal construction principles, representational formats, and learning mechanisms all differ from those of the human mind, they nonetheless generate highly structured linguistic behavior under certain conditions. By systematically manipulating training conditions, input distributions, and structural constraints of LLMs, we can observe the extent to which the extracted Galilean core principles hold only in highly idealized situations, and the extent to which they are robust to real-world perturbations. If a principle systematically collapses under slight distributional shifts, simplified training, or model degradation, this does not immediately refute the theory, but constitutes a significant methodological warning: the theory is more likely a modeling tool than a cognitive truth. Conversely, if certain principles recurrently emerge across multiple heterogeneous implementation

paths, the theory gains a new evidential basis for claiming their necessity, thereby enhancing its cognitive reality.

Here, the methodological practice aligns with Krakauer’s (2024) distinction between effective theories and fundamental laws in complex adaptive systems: linguistic principles operate as coarse-grained effective theories that abstract high-dimensional microscopic interactions into parsimonious macroscopic regularities, while LLMs function as heterogeneous experimental platforms that expose the broken symmetries and path-dependent frozen accidents embedded in real-world linguistic dynamics. Unlike conventional empirical falsification that targets axioms directly, this stress-testing probes the scope of validity of idealized effective theories by confronting them with non-equilibrium, history-dependent system behaviors that Galilean abstraction deliberately brackets.

This positioning leads us to further methodological discussion: whether research on LLMs will force Galilean generative theory to reconfigure its idealizations. The task of generative theory is to find the core computational constraints of language, but the problem is that the “core” is not a self-evident concept; it is always constructed through a set of idealization practices. When LLMs continually demonstrate that factors long ignored by theory (such as input magnitude, training trajectory, representational redundancy, distributional structure) can decisively influence linguistic behavior, can generative theory still unconditionally relegate these elements to non-linguistic mechanisms? In other words, must idealization in the age of LLMs be understood as a revisable strategy, a theoretical object open to reflection? As Weisberg (2013) pointed out, idealization in science is not simply the omission of details, but a purposeful selection of structure. When new empirical systems repeatedly reveal that ignored elements are functionally decisive, the boundaries of idealization must be re-evaluated. In this sense, LLMs constitute a methodological mirror for generative theory; they achieve empirical success by completely refusing idealization, thereby forcing the theory to ask which abstractions constitute genuine explanatory gains, and which are likely historical habits.

In summary, I propose that for generative theory to maintain its scientific legitimacy in the era of LLMs, it must insist on arguing for abstraction and avoid defending abstraction. Defending abstraction refers to the tendency of theorists, when faced with the complexity of empirical material or the success of statistical models, to insist on the principle that their subject matter is unobservable and can only be approached abstractly. Consequently, even when real systems exhibit phenomena inconsistent with theoretical predictions, these phenomena can be explained away as performance factors or non-essential interference. Arguing for abstraction means arguing why precisely certain distilled principles survive, rather than their variants; why certain dimensions should still be considered central or peripheral to a theory. The lesson LLMs bring to Galilean linguistics lies precisely in testing whether an idealization is a necessity of mathematical modeling, a cognitive necessity, or both.

5. Conclusion

The competition and dialogue between generative linguistics and LLMs constitute a contemporary

reverberation of the age-old rationalist-empiricist debate in the philosophy of science. I respond to my three questions. First, against Piantadosi (2024), I argue that the accidental truth of the empirical success of LLMs can't supplant the inquiry into the necessary truth of the constitutive principles of our language faculty, a distinction rooted in modal epistemology. Second, building upon Collins (2023), I propose operationalizing the domain of idealization, extra-domain necessary subsystems, and explicit interfaces to delineate resilience boundaries while preserving the explanatory core of theory. To make such operationalization practically usable, I also advance a minimum-description-length-oriented meta-theoretical heuristic and constrained consistency testing as complementary tools for theory assessment. Third, moving beyond mere coexistence, I frame LLMs as vehicles for theoretical stress-testing. By deploying heterogeneous computational implementations, we can probe the necessity-based claims of generative linguistics, discriminating real cognitive constraints from artefacts stemming solely from formal choices, and calibrating the reasonable boundaries of its Galilean idealization.

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